Son of Surya

GUIDE
R2
FUNDAMENTALS

FREE FOUNDATION GUIDE | BEGINNER

How to Read Your Electricity Bill

Units, slabs, fixed charges, sanctioned load and net metering — turn your DISCOM bill into the numbers that size your solar and predict your real savings.

kWh	kW	₹/unit	MF	1φ/3φ
UNITS	LOAD	SLABS	METER	PHASE

"Understand your solar before you buy your solar."

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R2 About This Guide

Guide	**R2 — How to Read Your Electricity Bill**
Category · Level · Price	Solar Fundamentals · Beginner · **Free**
For	Anyone in India who pays an electricity bill and is thinking about solar — homes, farms, shops
Prerequisites	R1 Electricity Basics (kW vs kWh)
Read next	R3 Sun, Site & Shade · SOS-01 Solar Starter · SOS-08 Net Metering

The problem

Your bill holds every number needed to size solar and estimate savings — but it is printed in codes, abbreviations and slabs that few people read beyond "Amount payable".

The knowledge gap

People confuse **sanctioned load (kW)** with **consumption (kWh)**, look only at one month, and assume every unit costs the same — when slab tariffs make the last units the most expensive.

The decision

After reading, you can extract your yearly consumption, your real cost per unit, your phase and sanctioned load — and judge what solar can and cannot remove from your bill.

The action

Collect 12 months of bills, fill in the bill worksheet in Section 13, and take it to any solar conversation.

01 What You Will Learn

- [ ] Where your consumer number, category, phase and sanctioned load are printed — and why each matters for solar
- [ ] How meter readings become units, including the multiplying factor
- [ ] How fixed charges, slab energy charges, adjustments and duty build up your total
- [ ] The difference between your **average** cost per unit and your **marginal** (last-unit) cost — and why solar saves at the marginal rate
- [ ] How to build a 12-month consumption profile and spot billing errors
- [ ] What a net-metered bill looks like after solar

02 Why This Matters

Every honest solar estimate starts from your bill. Size comes from your **units**; savings come from your **tariff**; what is allowed often depends on your **category and sanctioned load**. A vendor who sizes your system without asking

for your bills is guessing. A buyer who cannot read the bill cannot check the guess.

WHAT CHANGES IN INDIA?

Electricity tariffs are set separately by each **State Electricity Regulatory Commission** (SERC) through tariff orders, usually revised every year. So bill formats, slab limits, rates and charge names differ by state and DISCOM. This guide teaches the **structure** common to Indian bills; always read your own DISCOM's tariff for the numbers.

03 Fundamentals

Term	Simple meaning	Why it matters for solar
DISCOM	The company that supplies and bills your electricity	Approves rooftop connection and net meter
Consumer number	Your account ID with the DISCOM	Needed for every solar application
Category / tariff code	Residential, commercial, agricultural, industrial; LT (low tension) or HT (high tension)	Decides your rates, and which solar schemes and rules apply
Sanctioned (contract) load	Maximum power your connection is approved for — in kW or kVA	Fixed charges often depend on it; many DISCOMs link allowed solar capacity to it
Phase	Single-phase or three-phase supply	Your inverter must match
Units (kWh)	Energy you used in the billing period	The basis of system size
Billing cycle	Monthly or bi-monthly, depending on DISCOM	Convert bi-monthly bills to monthly before comparing

SIMPLE

Your bill = a small fixed amount for being connected + a charge for every unit you use + adjustments and taxes.

TECHNICAL

Most Indian retail tariffs are **two-part**: a fixed/demand charge (per kW, per kVA or per connection) and an energy charge (₹/kWh) in consumption slabs, plus regulatory adjustments and state electricity duty.

PRACTICAL

Solar reduces the **units** you buy. It does not normally reduce the fixed charge, and it reduces duty only on the part of the bill it shrinks.

DECISION

Value each solar unit at your **marginal** rate (the slab your last units fall in), not the bill total divided by units.

04 How a Bill Is Built



Figure 1. The building blocks of a typical Indian bill. Names differ by DISCOM — e.g. fixed charge, demand charge, FAC, FPPCA, electricity duty, meter rent, wheeling charge.

Slabs: why the last units cost the most

Most residential tariffs are **telescopic**: the first block of units is charged at the lowest rate, the next block at a higher rate, and so on. Only the units *inside* each block pay that block's rate. Some tariffs are non-telescopic — crossing a limit moves all units to a higher rate. Your tariff order says which applies.

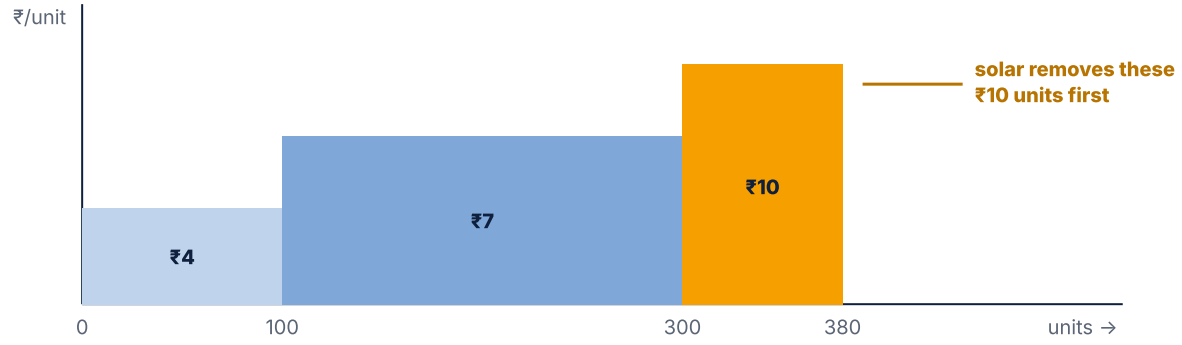


Figure 2. ILLUSTRATIVE EXAMPLE Telescopic slabs with invented rates. The area of each block is the energy charge for that block. Solar trims the top, most expensive block first.

05 Reading Your Bill — Line by Line

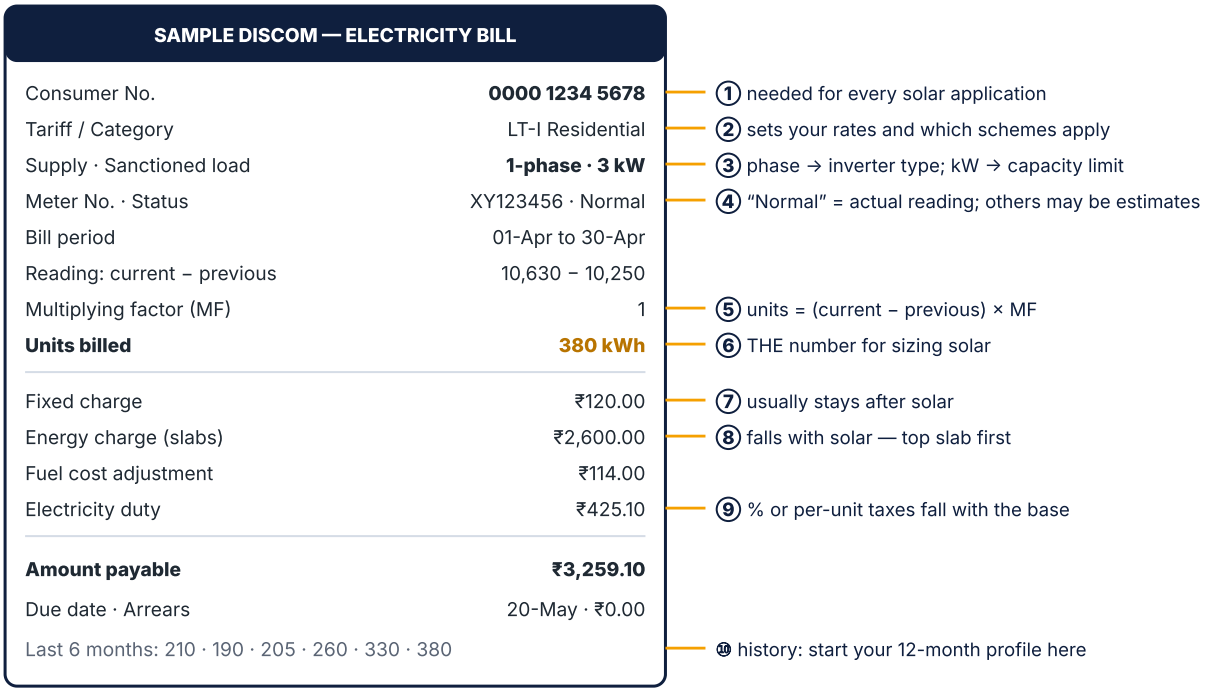


Figure 3. ILLUSTRATIVE EXAMPLE An invented bill using the slab rates of Figure 2 (₹4 / ₹7 / ₹10), fixed charge ₹120, adjustment ₹0.30/unit and duty 15%. Not any real DISCOM's format or tariff.

COMMON MISTAKE

Reading "Sanctioned load 3 kW" as "I use 3 units". Sanctioned load is a **power limit** (kW). Units billed is **energy** (kWh). They answer different questions.

06 Indian Context

Feature	What to know
Tariff set by state	Each SERC issues tariff orders; rates and slabs change, often yearly. Your DISCOM website publishes the current tariff.
Monthly vs bi-monthly	Some DISCOMs bill every two months. Slab limits may then apply per two months — check before comparing.
Adjustments	Fuel or power-purchase cost adjustment (FAC/FPPCA) varies month to month and is charged per unit.
State subsidies	Some states subsidise or waive units for eligible households; your bill may show a government subsidy line. Solar may change how much subsidy you receive — ask your DISCOM.
Agricultural supply	Farm connections may be metered or flat-rate (per HP) depending on state; flat-rate bills show HP, not units.

HT / commercial	Larger consumers may be billed on kVA demand, kVAh and time-of-day (TOD) rates. These need specialist analysis (SOS-14).
Your rights	The Electricity (Rights of Consumers) Rules, 2020 set national standards for metering, billing and complaints; unresolved complaints go to your DISCOM's Consumer Grievance Redressal Forum and then the Electricity Ombudsman.

After solar: the net-metered bill

With net metering, the meter records **import** (from grid) and **export** (to grid) separately. The bill shows both, the net units, and any credited or carried-forward units. How surplus is settled — carried forward, paid at a set rate, or lapsed at year-end — depends on your state's regulations (SOS-08).

Net-meter line	Units
Import (grid → home)	210
Export (home → grid)	160
Net billed (import – export)	**50**

ILLUSTRATIVE EXAMPLE. Note: export is only your *surplus*. Solar units used directly at home never reach the meter — so "import – export" is not your total solar generation. Check total generation on the inverter.

07 Practical Application — Build Your 12-Month Profile

1. Collect 12 consecutive months of bills (6 if bi-monthly). DISCOM apps and websites usually show bill history.
2. For each month write: units, amount, and the meter status (actual or estimated).
3. Add the units: **yearly consumption**. Divide by 365: **daily average**.
4. Note the highest and lowest months — the seasonal swing.
5. Find your **marginal rate**: the energy rate of the highest slab you usually reach, plus per-unit adjustments, plus duty.
6. Write down category, phase and sanctioned load.

SPOTTING BILLING PROBLEMS

- Status "average", "estimated", "locked" or "faulty" several months in a row — the bill may not reflect real use. Ask for an actual reading.
- A sudden jump with no change in habits — compare readings with the meter display yourself.
- Name on the bill differs from the owner — solar applications usually require the connection in the applicant's name. Ask your DISCOM about a name change first.

08 Calculations

C1 Units from meter readings

FORMULA	`Units = (Current − Previous) × MF`
WORKING	(10,630 − 10,250) × 1 = **380 units**. For a CT-operated meter with MF 20: a 19-unit difference = 19 × 20 = 380 units.
INTERPRETATION	Always check the MF on larger connections — forgetting it under-counts use 20× in that example.

C2 Building the bill (telescopic slabs)

ASSUMPTIONS	**ILLUSTRATIVE EXAMPLE** 0–100 @ ₹4 · 101–300 @ ₹7 · above 300 @ ₹10 · fixed ₹120 · adjustment ₹0.30/unit · duty 15% on all charges
WORKING	Energy: 100 × 4 + 200 × 7 + 80 × 10 = 400 + 1,400 + 800 = ₹2,600. Adjustment: 380 × 0.30 = ₹114. Subtotal: 120 + 2,600 + 114 = ₹2,834. Duty: 15% = ₹425.10. **Total ₹3,259.10.**
RESULT	**₹3,259.10 for 380 units**

C3 Average vs marginal cost per unit

FORMULAS	`Average = Total ÷ Units` · `Marginal = (top slab rate + per-unit adjustment) × (1 + duty)`
WORKING	Average: 3,259.10 ÷ 380 = **₹8.58**. Marginal: (10 + 0.30) × 1.15 = **₹11.85**.
INTERPRETATION	The last units cost almost 40% more than the average. Solar's first units are worth the marginal rate.

C4 What 150 solar units remove from this bill

WORKING	New consumption 230 units. Energy: 400 + 130 × 7 = ₹1,310. Adjustment: 230 × 0.30 = ₹69. Subtotal: 120 + 1,310 + 69 = ₹1,499. Duty: ₹224.85. **New total ₹1,723.85.**
RESULT	**Saving ₹1,535.25 = ₹10.24 per solar unit (vs ₹8.58 average)**
INTERPRETATION	Using the average rate would have *under*-estimated savings by about 16%. But as solar pushes you into lower slabs, each extra unit saves less — the next 100 units would save about ₹8.40 each ((7 + 0.30) × 1.15), and units below 100 only about ₹4.95. That is one reason very large systems give diminishing returns.

C5 From bills to a first solar size	
WORKING	Yearly units 3,300 → 9.04/day. In Maharashtra (planning figure 3.6 units/kWp/day, see R1/R3): 9.04 ÷ 3.6 = **2.5 kWp**.
INTERPRETATION	A first estimate only. Roof, shade, sanctioned load and DISCOM rules decide the final size.

09 Indian Case Study

ILLUSTRATIVE EXAMPLE — not a real customer, DISCOM or tariff

A family in Nagpur was told by a vendor: "Your bill is ₹3,200, so 4 kW will save ₹3,200 a month." Using this guide they found:

- Their 12 bills added up to 3,050 units — about 8.4 units/day → roughly **2.3 kWp** at Maharashtra's planning figure (8.4 ÷ 3.6).
- Their bill included a ₹120 fixed charge that solar would not remove, so ₹3,200 of savings was impossible.
- Their meter showed "average" status for three summer months — the true summer peak was unknown. They asked the DISCOM for an actual reading before finalising size.
- Their sanctioned load was 2 kW. They asked the DISCOM what capacity they could connect and whether a load change was needed.

OUTCOME

They asked vendors for quotes sized from their actual 12-month profile, with yearly kWh and savings calculated at their own slab rates. Promises based on the bill *amount* were set aside.

10 Cost & Economics

- **Savings are slab-dependent.** The first solar units remove your most expensive units; later units save less.
- **Fixed charges stay.** A zero bill is unusual even with large systems.
- **Export value is a state rule.** Units exported may be netted, paid at a regulated rate, or lapse — read SOS-08 before oversizing.
- **Tariffs change.** Savings estimates are only as good as the tariff year they use. Ask which tariff order an estimate is based on.

11 Common Mistakes

Mistake	Reality
Sizing solar from the bill amount in ₹	Size from units (kWh); the ₹ includes fixed charges and taxes
Using one month	Use 12 months — seasons change use a lot

Valuing every unit at the average rate	Use the marginal rate for the first solar units
Confusing sanctioned load with consumption	kW limit vs kWh used
Ignoring the multiplying factor	Units = reading difference × MF
Comparing a bi-monthly bill with a monthly one	Halve bi-monthly units first (and check slab periods)
Reading import – export as solar generation	Self-consumed solar never reaches the meter
Trusting estimated readings	Ask for actual readings before sizing

12 Buyer Protection

- [ ] Give vendors **12 months** of bills and ask them to show how they used them
- [ ] Ask: “Which tariff and slab did you use to calculate my savings?”
- [ ] Ask: “Did you include my fixed charge and any state subsidy I receive?”
- [ ] Ask: “How will exported units be credited in my state and category?”
- [ ] Check the connection is in your name and the category is correct before applying
- [ ] Be cautious of savings quoted as “your whole bill” or “zero bill”

13 Practical Checklist — Bill Worksheet

Month	Units	Amount ₹	Status	Month	Units	Amount ₹	Status
1				7			
2				8			
3				9			
4				10			
5				11			
6				12			

- [ ] Yearly units: _____ · Daily average: _____ · Highest / lowest month: _____ / _____
- [ ] Category: _____ · Phase: _____ · Sanctioned load: _____ kW
- [ ] Top slab rate ₹_____ + adjustment ₹_____ → marginal rate ₹_____ (with duty)
- [ ] Fixed charge per month: ₹_____ · State subsidy line? Yes / No
- [ ] Regional planning figure (R3): _____ → first size estimate: _____ kWp

SON OF SURYA FREE TOOL

Enter this worksheet into the **Son of Surya Solar Calculator** at sonofsurya.com to see size and savings with your own slabs — every assumption shown.

14 Problem Solving

Level	Problem	Solution
EASY	Readings 8,420 and 8,655, MF 1. Units?	8,655 − 8,420 = **235 units**
INTERMEDIATE	Using C2's illustrative tariff, what is the bill for 250 units?	Energy 400 + 150 × 7 = 1,450; adjustment 75; subtotal 120 + 1,450 + 75 = 1,645; duty 246.75 → **₹1,891.75** (average ₹7.57/unit)
REAL-WORLD	Bi-monthly bills total 4,200 units/year. Planning figure 3.4 (Tamil Nadu). First size? And what should you ask about slabs?	4,200 ÷ 365 = 11.5/day ÷ 3.4 = **3.4 kWp**. Ask whether slab limits apply per two-month period, and how surplus is settled under the state's current regulations.

15 FAQ

Q1 Where is my consumer number?
Usually near the top of the bill, sometimes called account or service number.

Q2 Why did my bill rise when units didn't?
Adjustment charges, tariff revisions or arrears can change. Compare line by line.

Q3 Will solar remove my fixed charge?
Normally no — it is linked to the connection or load, not units.

Q4 Should I reduce my sanctioned load?
Only after advice from your DISCOM — it can affect fixed charges and the solar capacity allowed.

Q5 Where are current tariffs published?
On your DISCOM's and SERC's websites, in the tariff order.

Q6 My bill is in someone else's name. Can I get solar?
Usually the connection must be in the applicant's name. Ask the DISCOM about the transfer process.

Q7 What is kVAh billing?
Billing on apparent energy, used for some larger consumers. Poor power factor raises the bill.

Q8 What is time-of-day (TOD) tariff?
Different rates at different hours, for some categories. It changes the value of solar units.

Q9 Is my free-units scheme affected by solar?
Possibly. Ask your DISCOM how solar interacts with any state subsidy.

Q10 I disagree with my bill. What now?
Complain to the DISCOM first, then the Consumer Grievance Redressal Forum, then the Ombudsman.

Q11 How many bills do I need for solar?
Twelve months gives the full seasonal picture.

Q12 Why is export lower than my solar generation?
Solar used at home is never exported or metered.

16 Glossary

Arrears — unpaid amount from earlier bills.

Billing cycle — period covered by one bill.

CGRF — Consumer Grievance Redressal Forum.

Consumer category — residential, commercial, agricultural etc.

Electricity duty — state tax on electricity.

Energy charge — ₹ per unit consumed.

FAC / FPPCA — fuel / power-purchase cost adjustment.

Fixed / demand charge — monthly charge for the connection or load.

HT / LT — high / low tension supply.

Import / export — energy from / to the grid.

kVAh — apparent energy billing unit.

Marginal rate — cost of your last (most expensive) unit.

MF — multiplying factor for meter readings.

Net metering — settling import against export.

Ombudsman — appeal authority above the CGRF.

Sanctioned load — approved maximum power.

SERC — State Electricity Regulatory Commission.

Slab — block of units charged at one rate.

Telescopic tariff — each slab charged only on units inside it.

TOD — time-of-day tariff.

17 What To Do Next

R1 Basics ✓ → R2 Your Bill ✓ → R3 Sun & Site → SOS-01 Starter → Free Calculator

18 Sources

Source	Used for	Checked
Electricity Act, 2003 (tariff determination by State Commissions; consumer grievance forum, Sec. 42)	Who sets tariffs; CGRF and Ombudsman	reference
Ministry of Power — Electricity (Rights of Consumers) Rules, 2020	Consumer rights on metering, billing, complaints	reference
MNRE Rooftop Solar FAQ via PIB (28 Apr 2022)	Net-meter function	7 Oct 2026
Son of Surya regional planning figures (see R1/R3)	Size estimates C5, problems	7 Oct 2026
Your DISCOM's current tariff order	Your actual slabs and rates	check yourself

19 Disclaimer

Published by Son of Surya for general education; not legal, financial or engineering advice. All tariffs, bills and families in this guide are ILLUSTRATIVE EXAMPLES and do not represent any real DISCOM, tariff order or customer. Tariff structures, slab limits, subsidies and net-metering rules differ by state and change over time — confirm with your DISCOM and State Commission. Savings estimates are not guarantees.

Son of Surya

India’s structured solar knowledge library

Free foundation guides, practical calculators and honest, step-by-step knowledge — so every Indian home, farm and business can understand solar before buying it.

UNDERSTAND → CALCULATE → VERIFY → COMPARE → DECIDE → ACT

FREE TOOL

Son of Surya Solar Calculator

Enter your bills and location. See system size, yearly units and savings — with every assumption shown.

FREE TOOL

Solar Project Passport

Keep your solar project’s key numbers, documents and warranties in one place. An engineer-verified version is available.

FREE FOUNDATION GUIDES

- **R1** Electricity Basics for Solar
- **R2** How to Read Your Electricity Bill
- **R3** Sun, Site & Shade
- **SOS-01** Solar Starter Guide
- **R11** Solar Dictionary (English–Hindi)

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Powering the land. Protecting the water. Growing the future.

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